Tangent Lévy v2
Risk-Neutral Drift Audit Summary

quantdata / Dysanos

9 October 2026

Abstract

This note records the final v2 audit of a reduced Tangent Lévy construction under the risk-neutral measure Q. The experiment uses the native Tangent Lévy short-end spot compensator, a deterministic Musiela shift, and a restricted universe consisting of spot plus 44 fixed-cash-strike options. The two conditional drift tests do not reject the Q martingale restriction. The literal coordinate-wise unconditional Gaussian diagnostic reports many extreme statistics, but those statistics are dominated by compound-Poisson coordinates whose compensating jumps are too rare to be sampled by 100,000 paths. The result is therefore evidence about this reduced Q construction, not a successful calibration of the full stochastic model in the paper.

1 Scope and implementation

The population contains 100,000 simulated paths (50,000 antithetic pairs), initialized from the 1 June 2022 SPX surface. The tested trading universe is spot plus 44 options: eleven normalized-moneyness instruments at each of 5, 10, 15, and 20 business-day maturities. Strikes remain fixed in cash units over each one-step return. Spot is advanced with the Tangent Lévy codebook's short-end compensator, while the codebook is transported by deterministic Musiela shift substeps.

The implementation includes the finite-activity no-jump atom in transform pricing, lattice-consistent FFT pricing, removal of numerical FFT dust, tail cutoffs under both cash and share intensities, short-end midpoint evaluation in Musiela substeps, float64 audit returns, and exact fold-level entropy scoring. These details are necessary for the simulated spot and option prices to be internally consistent under Q.

2 Calibration limitation

The attempted historical reconstruction did not pass its mandatory gate on any of 335 dates. Across the history the maximum relative L^1 error was 1.3817 and the maximum repricing RMSE was 525 quoted half-spreads. For the retained 1 June 2022 initialization, the relative L^1 error is 0.9473, the repricing RMSE is 20.185 half-spreads, and the maximum absolute normalized price error is 0.00977. This confirms the expected poor fit.

Because the reconstructed history supplied no positive covariance mode, the final audit uses zero stochastic factors. It is a deterministic-codebook Tangent Lévy model, not a fitted version of the paper's full stochastic factor/HJM specification. The drift-test methodology is adapted from `paper/main2.tex`; that paper's underlying SDLV experiment uses a Gaussian/information-bridge spot construction and does not by itself calibrate the sampling distribution of a rare-event compound-Poisson diagnostic.

Test	Mean score	Pair SE	t statistic	95% interval	Result
Conditional, squared	3.340×10^{-8}	7.299×10^{-8}	0.458	$[-1.096, 1.764] \times 10^{-7}$	**PASS**
Conditional, entropy	1.702×10^{-9}	2.915×10^{-9}	0.584	$[-4.012, 7.415] \times 10^{-9}$	**PASS**

Table 1: Cross-fitted conditional drift tests on the retained population.

3 Final Q results

Both conditional confidence intervals contain zero and both absolute t statistics are below 1.96. Accordingly, neither conditional test rejects the null of zero predictable drift at the 5% two-sided level.

The literal unconditional report tests 570 time–instrument coordinates. It finds a maximum absolute z statistic of 1.485×10^9, a 99th percentile of 1.196×10^8, 204 coordinates above 1.96, 150 above 2.576, and 138 above 5. Read literally as Gaussian tests, these coordinates fail.

That literal reading is not statistically valid for the problematic coordinates. Some reconstructed annual intensities are approximately 1.5×10^{-6}, corresponding to a one-day jump probability near 5.95×10^{-9}. A 100,000-path population is therefore overwhelmingly likely to contain no compensating jump. The empirical variance collapses while the analytically compensated mean remains nonzero, producing enormous studentized ratios. On the 432 coordinates with pair standard error exceeding 10^{-12}, the maximum absolute statistic is 3.597, the 99th percentile is 3.551, 66 exceed 1.96, 12 exceed 2.576, and none exceeds 5. This filtered view is diagnostic only; it is not a replacement preregistered test.

4 Conclusion

The strongest defensible conclusion is narrow. The two requested conditional tests find no detectable predictable drift under the reduced Q simulation. The unconditional Gaussian coordinate test cannot confirm the same claim at this Monte Carlo size because the fitted finite-activity model contains events far rarer than the population can resolve. Its extreme statistics are a rare-event resolution failure, not direct evidence of an exploitable arbitrage. A definitive unconditional assessment would require analytic conditional moments, importance sampling, or a test explicitly designed for the compound-Poisson law.

This audit does not validate the market fit, the physical-measure dynamics, or the omitted stochastic codebook factors. It establishes that the implemented reduced construction uses the Tangent Lévy compensator consistently and that its two cross-fitted conditional drift tests pass on the retained population.

5 Retained artifacts

The canonical result root is `/home/hans/quantdata_data/training/dysanos/tangent-levy/v2/spx-891-v2`. It retains the 1 June 2022 initialization checkpoint, the `q-reduced-population` arrays, the unconditional report `q-reduced-audit.json`, both conditional JSON reports, and their current cross-fit checkpoints. Superseded daily reconstruction checkpoints and legacy aggregate calibration logs are intentionally excluded.